Project Report: Analysis of a Telecom Company's Tariff Plans

1. Project Objective

The objective of the project is to analyze two tariff plans offered by a telecommunications company, determine which plan generates higher revenue, and assess whether the observed difference is statistically significant.

The analysis results are intended to assist the business in making decisions regarding the future development of its tariff lineup and in identifying which plan offers greater commercial value to the company.

2. Business Objective

The company offers clients two tariff plans:

"Smart"

"Ultra"

The tasks are to:

Examine the features of the tariff plans.

Analyze user behavior for both plans.

Calculate monthly revenue per client.

Compare the average revenue from users of both plans.

Test the statistical significance of the differences.

Formulate a business recommendation based on the analysis results.

Key business question:

Does the average monthly revenue from "Smart" and "Ultra" users differ, and which plan is more profitable for the company?

3. Tariff Description

Information regarding the terms of two tariff plans was used for the analysis.

"Smart" Tariff

monthly fee: 550 UAH ;

includes 500 minutes of talk time;

includes 50 SMS;

includes 15 GB of data.

If the included allowance is exceeded, the customer pays an additional fee for the services used.

Cost of services exceeding the tariff package: 1. minute of conversation: 3 UAH; 2. message: 3 UAH; 3. 1 GB of internet data: 200 UAH.

"Smart" Tariff

monthly fee: 1950 UAH;

includes 3000 minutes of talk time;

includes 1000 SMS;

includes 30 GB of data.

Cost of services exceeding the tariff package: 1. minute of conversation: 1 UAH; 2. message: 1 UAH; 3. 1 GB of internet data: 150 UAH.

4. Data for analysis

The analysis uses data on users and their activity:

customer information calls.csv;

user tariff plan tariffs.csv;

date and number of messages sent messages.csv;

date and volume of internet data used internet.csv;

date and duration of calls calls.csv;

5. Data Preparation

Data preprocessing was required before conducting the analysis.

Key steps:

Checking data types.

Handling missing values.

Checking for duplicates.

Converting dates to the correct format.

Correction of typos and errors.

Calculation every month service usage.

calls per month

	user_id	year_month	minutes	calls
0	1000	2018-05	159.0	22
1	1000	2018-06	172.0	43
2	1000	2018-07	340.0	47
3	1000	2018-08	408.0	52
4	1000	2018-09	466.0	58

messages per month

	user_id	year_month	messeges
0	1000	2018-05	22
1	1000	2018-06	60
2	1000	2018-07	75
3	1000	2018-08	81
4	1000	2018-09	57

internet per month

	user_id	year_month	gb_used
0	1000	2018-05	3.0
1	1000	2018-06	23.0
2	1000	2018-07	14.0
3	1000	2018-08	14.0
4	1000	2018-09	15.0

6.Calculation of the average actual volume of service consumption by users

avg actual volume of service

	user_id	tariff	avg_minutes	avg_messeges	avg_gb_used
0	1000	ultra	320.75	62.0	13.62
1	1002	smart	216.71	10.0	16.71
2	1003	ultra	764.40	76.0	11.80
3	1004	ultra	123.38	149.0	18.75
4	1005	smart	550.42	46.0	8.58

Calculation of average consumption of minutes, SMS, and internet data.

avg users needs

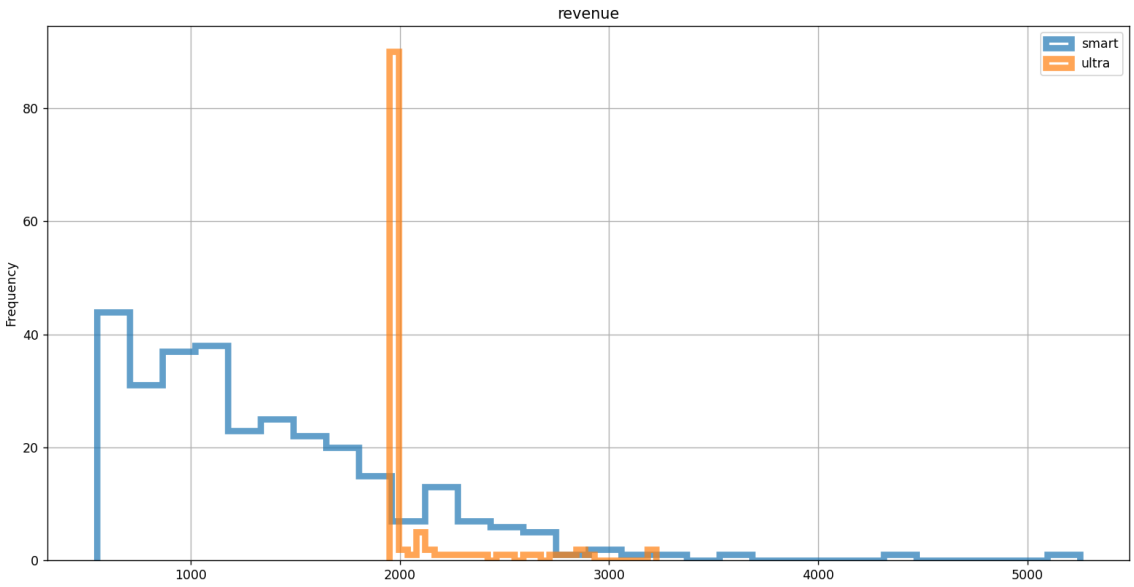
	tariff	avg_minutes	avg_messeges	avg_gb_used	std_minutes	std_messeges	std_gb_used
0	smart	424.94	37.64	16.46	148.05	24.88	3.78
1	ultra	559.16	65.42	19.59	274.76	42.41	7.87

7.Revenue general information:

revenue general information

tariff	user_num	total_sum	mean_rev	median_rev
smart	301	2471648.0	1339.89	1189.8
ultra	118	1540800.0	2066.66	1950.0

Revenue distribution.



8. Comparison of tariff revenues

The main indicator of the effectiveness of the tariff is the average monthly revenue per user.
Need to compare
H_0: the average revenue of Smart tariff users is equal to the average revenue of Ultra tariff users.
H_1: The average revenue of users of the two tariffs differs.
Level of statistical significance:
$\alpha = 0.05$.

TtestResult
pvalue= 6.91205648747052e-27

9. Interpretation of results

The result of a statistical test must be interpreted not only from a statistical perspective but also from a business perspective.
The average monthly revenue from users of the "Ultra" plan is statistically significantly higher than that from users of the "Smart" plan. It is highly probable that the observed difference is not due to random sampling fluctuations.

The difference is:

2066.66 − 1339.89 = 726,77 rubles per user per month.
The number of users is significant,so the potential impact on the business could be substantial.

9. Overrun

The rate plans include bundled allowances; we need to look at the following separately:
- how many users exceed their minute allowance;
- how many exceed their data allowance;
- how many exceed their SMS allowance.

tariff	overlim_minutes	overlim_messeges	overlim_gb_used
smart	34.85%	28.47%	58.31%
ultra	0.00%	0.00%	12.08%

10. Business Recommendations

"Ultra"plan yields a higher ARPU and greater average profit per user. "Smart" plan, meanwhile, has a larger user base and generates a comparable total revenue. A statistical test indicates that the difference in average revenue between the two plans is statistically significant ($p < 0.05$). Users of "Ultra" plan also exhibit higher mobile data consumption. Therefore, when evaluating the performance of these plans, it is essential to consider not only ARPU but also the size of the customer base, profitability, and consumption patterns.

The "Ultra" plan shows a statistically significantly higher average revenue; it is recommended to:

-continue promoting the "Ultra" plan among clients with high service usage;

-analyze the possibility of migrating some "Smart" plan users to "Ultra";

-identify customer segments with a high potential for switching;

-separately examine users who regularly exceed the limits of the "Smart" plan.

"Smart" plan has large customer base, the following should be considered:

-user retention;

-customer acquisition cost;

-tariff profitability;

-consumption patterns;